

A3Part2

September 22, 2025

```
[6]: from pyspark.sql import SparkSession
```

```
spark = SparkSession.builder \
    .appName("Test") \
    .master("local[*]") \
    .getOrCreate()
```

```
[7]: Titanic = spark.createDataFrame([("Children", "First", 6, 0), ("Children", "Second", 24, 0), ("Children", "Third", 27, 52), ("Men", "First", 57, 118), ("Men", "Second", 14, 154), ("Men", "Third", 75, 387), ("Men", "Crew", 192, 693), ("Women", "First", 140, 4), ("Women", "Second", 80, 13), ("Women", "Third", 76, 89), ("Women", "Crew", 20, 3)], ["sex", "class", "survived", "died"])
Titanic.show()
```

sex	class	survived	died
Children	First	6	0
Children	Second	24	0
Children	Third	27	52
Men	First	57	118
Men	Second	14	154
Men	Third	75	387
Men	Crew	192	693
Women	First	140	4
Women	Second	80	13
Women	Third	76	89
Women	Crew	20	3

```
[8]: Titanic.filter((Titanic["survived"] + Titanic["died"])*0.8 <= Titanic["survived"]).show()
```

sex	class	survived	died
Children	First	6	0

Children	Second	24	0
Women	First	140	4
Women	Second	80	13
Women	Crew	20	3
+-----+	+-----+	+-----+	+-----+